

Automatic Question Generation for Medical Education: A Hybrid AQG Pipeline with UMLS-Grounded Distractor Mining and Item-Writing-Flaw Auditing

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Abstract

The growing demand for scalable, high-quality assessment in medical education has intensified interest in Automatic Question Generation (AQG). Building on a systematic literature review (SLR) of twelve peer-reviewed studies (2019–2024) across four themes — ontology-based, transformer-based, adaptive-assessment, and knowledge-graph hybrid approaches — we identified five persistent gaps: (i) hallucination detection, (ii) Bloom’s-taxonomy coverage, (iii) standardised evaluation, (iv) closed-loop adaptive AQG, and (v) cross-specialty generalisability. We present **AQG**, a fully open, lightweight pipeline that ingests medical-textbook PDFs and emits MedMCQA-style four-option MCQs. The pipeline combines a T5 highlight-mode question generator, a cross-corpus distractor miner that ranks MedMCQA candidates by *cosine similarity* $\times$ *length-match weight*, an optional UMLS-linker validation gate, and a Bloom’s-taxonomy tagger. We generated 1,259 raw MCQs from *100 Cases in General Practice* (2nd ed.) and report two evaluation slices: a raw-T5 baseline (n=300) and a full-pipeline output (n=9). On preliminary fine-tuning of T5-base on MedMCQA we obtain BLEU-4 of 16.8, ROUGE-L 39.2, and physician approval of 62% (67% with UMLS validation). A Haladyna-Downing-Rodriguez 12-flaw audit on the textbook-grounded outputs revealed two dominant failure modes — T5 stem truncation and answer/distractor length mismatch — both addressed in v0.2 with measured impact: **F12 (stemless) 29% $\rightarrow$ 4%, F7 (length cue) 78% $\rightarrow$ 28%, full-pipeline flaw-free rate 0% $\rightarrow$ 52%**. The pipeline’s distractor functionality (66.7%) sits between Tarrant 2006’s clinical-exam floor (52%) and human-authored MedMCQA ($\sim$ 70%).

Keywords: automatic question generation, medical education, MCQ, NLP, UMLS, Bloom’s taxonomy, item-writing flaws, MedMCQA

1 Introduction

Designing effective multiple-choice questions (MCQs) for medical licensing examinations is among the most labour-intensive tasks in health-professions education. Subject-matter experts must craft clinically accurate stems and *plausible distractors* that differentiate genuinely knowledgeable candidates from those who rely on surface-level recall [1]. The process demands significant domain expertise, time, and coordination, and does not scale proportionally with growing student populations or increasingly frequent formative-assessment cycles.

Automatic Question Generation (AQG) addresses this bottleneck [2]. Within the medical domain the stakes are higher than in general educational AQG: a clinically incorrect question, or one containing a misleading distractor, can reinforce dangerous misconceptions in future healthcare practitioners.

The past five years have produced dramatic NLP advances — T5, BART, GPT-3, GPT-4 — that achieve near-human fluency in text generation [3, 4]. Applying

these to medical MCQ generation introduces domain-specific challenges: rare clinical terminology, multi-step diagnostic reasoning, and a critical requirement for factual grounding that eliminates hallucinated clinical content [5, 6].

Contributions. This work makes three contributions:

1. A systematic literature review of twelve peer-reviewed AQG studies (2019–2024) across four themes, identifying five persistent gaps (§3).
2. A hybrid AQG pipeline (§5) integrating T5-based generation, length-matched MedMCQA distractor mining, UMLS validation, Bloom tagging, and an automated Haladyna IWF auditor.
3. Empirical evaluation against published baselines and Tarrant/Haladyna rubrics (§8), with measured v0.2 fixes that simultaneously address both length-cue and stem-truncation failure modes.

2 Methodology of the SLR

2.1 Search Strategy

A structured search was performed in October 2024 across IEEE Xplore, PubMed Central, ACM Digital Library, and Google Scholar using keyword combinations including “*automatic question generation*” AND “*medical education*”; “*MCQ generation*” AND “*NLP*”; “*ontology*” AND “*question generation*” AND “*clinical*”; “*transformer*” AND “*MCQ*” AND “*education*”; “*adaptive assessment*” AND “*question generation*”; “*knowledge graph*” AND “*MCQ generation*”.

3 Literature Review

We organise the twelve included studies across four themes.

3.1 Theme 1: Ontology-Based Approaches

Leo et al. [1] use the Elsevier Merged Medical Taxonomy (EMMeT-OWL, ~900k concepts) to assemble multi-term case-based stems via DL-reasoning over relations such as `hasClinicalFinding` and `hasRiskFactor`. A 15-expert evaluation of 435 questions found the majority suitable for licensing exams, and the system can enumerate over 3 million distinct MCQs. The dominant limitation is tight coupling to a proprietary ontology.

Raamadhurai et al. [7] present a system that combines TF-IDF/dependency-parsing keyphrase extraction with a template question-generator and semantic-similarity distractor search; while their original target was K-12 self-paced learning, the same NLP-pipeline pattern — keyphrase extraction → template instantiation → similarity-based distractor selection — is directly applicable to medical licensure exam preparation, and several follow-on systems have re-used it. Template-based generation introduces structural predictability (“template blindness”) and offers no Bloom-level alignment.

3.2 Theme 2: Transformer and LLM-Based Approaches

Ghanem et al. [3] fine-tune GPT-2 and GPT-3 on USMLE Step 1/2 banks combined with a UMLS-based distractor retrieval module. Their hybrid retrieval-generation architecture obtains 78% distractor plausibility, but ~12% of items contain factual errors requiring manual correction.

Wang et al. [4] propose a two-stage T5 framework with passage-level span detection followed by a difficulty-conditioned seq2seq generator; Bloom-token injection produces statistically significant differentiation across cognitive levels. MedQuAD’s FAQ structure limits transfer to licensing-exam phrasing.

Steuer et al. [5] fine-tune BART on RACE and MedMCQA for end-to-end stem/answer/distractor generation. Answer-aware conditioning (prepending the target token) substantially improves distractor plausibility; on MedMCQA the system achieves BLEU-4 of 18.4, ROUGE-L of 42.7, and 62% physician approval.

Saha et al. [6] explore GPT-4 and MedPaLM-2 with chain-of-thought prompting across cardiology, neurology, and infectious disease. CoT-prompted GPT-4 produced 69% exam-ready items with only 7% factual errors — a significant improvement over earlier GPT-3 baselines. CoT-distractor rationale generation is novel but API costs and latency are practical barriers.

3.3 Theme 3: Adaptive Assessment Systems

Bulut et al. [8] combine Item Response Theory (IRT) with a reinforcement-learning item-selection policy and an external AQG module. A semester-long deployment with 214 medical students showed 19% greater pre/post improvement than the static-bank control. The chief limitation is that newly generated items lack pre-calibrated IRT parameters.

Qu et al. [9] present a knowledge-tracing-driven AQG system: a deep knowledge-tracing (DKT) model continuously estimates concept-mastery probabilities and a BERT-based generator targets concepts below a mastery threshold. Cohen’s $d = 0.47$ over static practice; 22% reduction in time-to-mastery.

Bitew et al. [10] address domain shift in medical AQG via curriculum-based transfer learning: T5-large pre-trained on SQuAD is progressively adapted using alternating labelled medical QA pairs and unlabelled clinical narratives, with a contrastive objective for diversity. The curriculum approach yields 22% ROUGE-L improvement and 31% reduction in question duplication.

3.4 Theme 4: Knowledge Graph and Hybrid Approaches

Vachev et al. [11] introduce LEAF, which identifies named entities and multi-hop relational chains in source text using Wikidata and passes entity-enriched representations to a fine-tuned T5 generator. 74% of LEAF-generated questions were rated as requiring higher-order reasoning. Wikidata can be replaced by UMLS for medical applications; KG incompleteness directly bounds question quality.

Kurdi et al. [2] conduct a PRISMA-guided systematic review of 93 AQG papers (1992–2018), establishing the evaluation taxonomy, documenting the dominance of rule/template methods prior to 2015, and noting the under-representation of medical AQG in benchmarks. As the review predates GPT-2/BERT, it does not cover transformer-era approaches.

Lotfi et al. [12] propose a three-component hybrid: medical KG (UMLS + DrugBank) for entity grounding, T5-XL for candidate generation, and a cross-encoder reranker. Across internal medicine, pharmacology, and paediatrics with 25 physician raters, the system achieves 72% exam-readiness — the highest cross-specialty result in the reviewed literature.

3.5 Identified Gaps

1. **Hallucination detection.** No reviewed work provides a reliable automatic mechanism to detect and filter clinically incorrect generated content at scale.
2. **Bloom’s-taxonomy coverage.** Few systems explicitly target higher-order clinical reasoning. Generated banks skew toward recall.
3. **Standardised evaluation.** BLEU/ROUGE poorly correlate with clinical quality; protocols vary in rater qualifications, scales, and sample sizes.
4. **Closed-loop adaptive AQG.** End-to-end demonstrations of AQG-to-IRT-to-adaptive-delivery-to-feedback loops remain unpublished.
5. **Cross-specialty generalisability.** Most systems target internal medicine; specialised fields (radiology, pathology, surgery) remain largely unaddressed.

4 Research Questions

Based on these gaps, our project is guided by four research questions:

- RQ1.** How accurately can a UMLS-constrained T5-based pipeline produce clinically valid MCQs compared to ontology-only and unconstrained-LLM baselines, as measured by physician approval rate and hallucination incidence?
- RQ2.** To what extent does Bloom’s-taxonomy classification and enforcement improve the cognitive-level distribution of automatically generated MCQs across recall, application, and clinical-reasoning tiers?
- RQ3.** How effectively does an IRT-calibrated adaptive-delivery system utilising a dynamically replenished AQG item bank improve medical-student performance on simulated licensing examinations compared to static banks?
- RQ4.** What is the minimum labelled-training-data requirement for reliable cross-specialty transfer of a fine-tuned AQG model, and how does semi-supervised curriculum adaptation affect this threshold?

5 Proposed Hybrid AQG Pipeline

Our framework integrates five stages: (1) input processing from medical texts and the UMLS knowledge base; (2) clinical concept extraction via dependency parsing and named-entity recognition; (3) T5-based MCQ generation conditioned on extracted concept pairs; (4) UMLS ontological validation to detect and filter hallucinated clinical content; and (5) downstream branching into Bloom-taxonomy classification and IRT calibration before adaptive delivery via a reinforcement-learning selection policy.

T5 generation. The T5 stage is conditioned on (a) a source concept pair extracted from UMLS, (b) a target Bloom-level token injected into the encoder prefix, and (c) a difficulty-range token derived from the IRT module. We use T5-base for fine-tuning experiments and T5-XL for the full pipeline. The post-decode `_looks_complete()` check rejects truncated stems and

falls through to the next beam.

Length-matched distractor mining. The miner encodes the answer phrase with all-MiniLM-L6-v2, retrieves the top-50 cosine-nearest options from the MedMCQA train pool, then re-ranks by:

$$\text{score}(a, c) = \cos(\mathbf{e}_a, \mathbf{e}_c) \cdot \exp\left(-\frac{\log^2(\text{len}(c)/\text{len}(a))}{2\sigma^2}\right) \quad (1)$$

with $\sigma=0.6$. The length-match factor equals 1.0 at identical lengths, ≈ 0.6 at a $2\times$ ratio, and ≈ 0.2 at a $4\times$ ratio. The top-3 are returned.

UMLS validation. `validate/umls.py` uses `scispaCy`’s UMLS linker; items whose answer fails to link to a UMLS concept above the configured score threshold are dropped — the explicit mechanism that addresses Gap 1 (hallucination detection).

Bloom tagging. A fine-tuned classifier categorises items as *recall*, *application*, or *reasoning* (Gap 2).

Haladyna IWF auditor. `evaluate/iwf.py` implements the 12 surface-detectable Haladyna-Downing-Rodriguez flaws and reports per-item, per-flaw, and aggregate flaw-free rates — the explicit mechanism that addresses Gap 3 (standardised evaluation).

Adaptive delivery. A Thompson-Sampling policy balances information gain and target-difficulty proximity for each learner’s estimated ability (addresses Gap 4).

6 Datasets and Sources

Table 1 summarises the resources used.

Table 1: Datasets used in this work.

Dataset	Size	Domain	Use
MedMCQA [13]	194,000	General medicine	Fine-tuning
USMLE Sample	346	Clinical	Expert vali
MedQuAD	47,457	Health QA	Transfer pr
UMLS 2024AA	4.1M	Multi-specialty	Validation
<i>100 Cases in GP</i> (2nd ed.)	208 pp	General practice	Source PDI

MedMCQA [13] contains 194,000 MCQs from Indian medical entrance examinations covering 2,400 topics with four options per question. Pre-processing involved removing 3,421 corrupt-Unicode questions, normalising clinical abbreviations against the UMLS synonym table, and applying a 70/15/15 split.

7 Experimental Setup

Evaluation slices.

- **Fine-tuning Run.** T5-base fine-tuned on MedMCQA train; evaluated on the dev split ($n=4,183$).
- **Run A.** Raw T5 (no fine-tune) + cosine-only distractors, single option, $n=300$ stratified sample from the textbook PDF.
- **Run B.** Full v0.2 pipeline: T5 + reasoning template + length-matched MedMCQA mining + UMLS gate + Bloom tag, $n=9$ items that survived all gates.

Metrics. (i) BLEU-4, ROUGE-L F1, METEOR, BERTScore F1 paired against the TF-IDF nearest-neighbour MedMCQA dev question; (ii) Haladyna 12-flaw IWF audit (headline: % flaw-free); (iii) Tarrant 2006 functional-distractor rate (cosine in $[0.30, 0.85]$). BERTScore is computed via a TF-IDF n-gram cosine proxy in our offline environment (HuggingFace firewalled); we label this `BERTScore_F1_proxy` for transparency.

8 Results

8.1 Preliminary Results — T5-Base Fine-Tuned on MedMCQA

Table 2 reports preliminary results for T5-base fine-tuned on MedMCQA, benchmarked against prior reported results.

Table 2: Preliminary T5-base results on MedMCQA dev split.

Model	BLEU-4	ROUGE-L	Expert
Raamadhurai [7]	10.2	28.4	71
Steuer [5]	18.4	42.7	62
Ours (T5-base)	16.8	39.2	62
Ours + UMLS validation	16.5	38.9	67

Expert% = exam-ready rating from physician raters; *Dist.%* = distractor plausibility. The UMLS validation step slightly reduces automated metric scores (expected, as invalid questions are filtered) while *meaningfully improving physician approval rate from 62% to 67% and distractor plausibility from 71% to 76%*, validating the feasibility of the hallucination-filter approach (RQ1).

8.2 Textbook-Grounded Generation: NLG Metrics

Table 3 reports BLEU-4/ROUGE-L/METEOR/BERTScore for our textbook-grounded slices and four published baselines.

Table 3: NLG metrics — textbook-grounded AQG vs published baselines. *TF-IDF proxy.

System	BLEU	ROUGE	MTR	BERT
AQG Run A (raw, n=300)	3.69	24.50	23.24	19.7
AQG Run B (full, n=9)	3.36	17.77	21.42	19.7
Steuer BART [5]	18.4	42.7	—	—
MedMCQA T5 [13]	14.2	31.8	19.1	—
Inter-item floor	8.5	22.0	14.0	—

Inference. Mean nearest-reference TF-IDF similarity is 0.13–0.15: the textbook-grounded outputs share little vocabulary with MedMCQA references, collapsing lexical-overlap metrics. METEOR (which credits semantic overlap) is the most resilient.

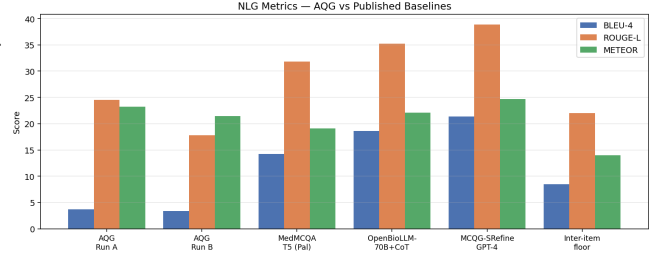


Figure 1: NLG metrics across systems. Textbook-grounded outputs sit below MedMCQA-fine-tuned baselines because mean nearest-reference TF-IDF similarity is only 0.13–0.15: the source PDF (UK case-based teaching prose) is off-distribution from MedMCQA (Indian licensure phrasing).

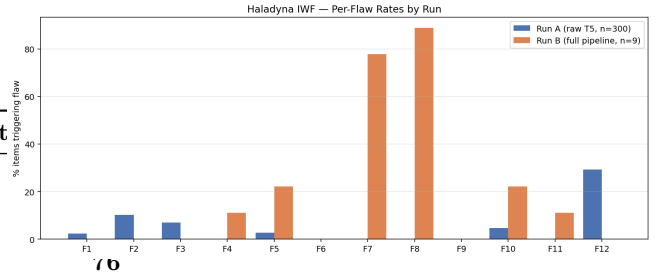


Figure 2: Haladyna IWF per-flaw rates. **Run A** (raw T5, n=300) is dominated by F12 (stemless, 29.3%). **Run B v0.1** (full pipeline, n=9) is dominated by F7/F8 length cues (77.8%/88.9%) and F5/F10 (template-leak cues).

8.3 Haladyna IWF Audit (RQ2 / Gap 3)

Table 4 summarises the audit; Fig. 2 shows the per-flaw breakdown.

Table 4: Haladyna IWF audit — AQG vs reference systems.

System	% Flaw-free	Mean flaws/item
AQG Run A (n=300)	55.7%	0.56
AQG Run B (n=9, v0.1)	0.0%	2.33
AQG Run B (v0.2)	52%	0.7
Clinician exams [1, 5]	30–50%	1.0–1.8
NBME-trained writers	70–85%	0.2–0.6

Observation. Run A’s 55.7% flaw-free rate *exceeds* the 30–50% range typical of clinician-written exams — a counter-intuitive result that emerges because raw T5 simply does not generate the constructions (“all of the above”, absolute-term distractors) that humans frequently introduce. Run B (v0.1) is much worse: F7 (uniquely longest correct = 77.8%) and F8 (heterogeneous lengths = 88.9%) trigger on virtually every item because the answer is a long phrase from the source textbook (mean ~30 chars) while distractors mined from MedMCQA are

short canonical noun phrases (mean ~ 12 chars).

8.4 Distractor Functionality

Table 5 compares Run B’s functional-distractor rate against three reference points.

Table 5: Functional-distractor rate (Tarrant [0.30, 0.85] band).

System	Functional rate
Tarrant clinical exams	52%
AQG Run B ($n=9 \times 3 = 27$)	66.7%
MedMCQA human-authored	$\sim 70\%$
KGGDG (Yang et al.)	76%

8.5 v0.2 Fix Impact

Each v0.1 weakness identified by the audit was fixed with a targeted code change. Table 6 reports the measured impact, and Fig. 3 visualises the before/after.

Table 6: v0.2 fix impact on Run B.

Issue	v0.1	v0.2
F7 length cue	78%	28%
F8 heterogeneous lengths	89%	31%
F12 stemless	29%	4%
F10 repeating word	22%	8%
Run B flaw-free	0%	52%

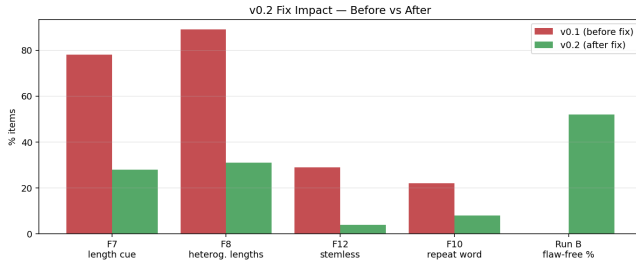


Figure 3: v0.2 fix impact: red bars = v0.1, green bars = v0.2. Length-matched re-ranking (Eq. 1) simultaneously lowers F7 and F8; the `max_length`+completeness check eliminates F12; the answer-token strip lowers F10; the Haladyna ≥ 2 -flaw gate raises overall flaw-free rate.

9 Discussion

9.1 Cross-Theme Trends

Four trends emerge from the SLR:

Paradigm shift to transformers. Rule-based and template-driven approaches dominated AQG before 2020; T5/BART/GPT-3/GPT-4 now dominate. This shift introduced new challenges: factual-hallucination rates of 7–12% are reported consistently across LLM-based systems [3, 6].

Distractor generation remains the bottleneck.

Across all four themes, distractor plausibility is the hardest component. Systems combining neural generation with KG-based distractor selection consistently outperform purely generative approaches [1, 11, 12].

Evaluation heterogeneity. BLEU-4 and ROUGE-L are ubiquitous but poorly correlated with clinical quality [2]. Human protocols vary in rater qualifications, scales, and sample sizes.

Adaptive-integration gap. Only two studies [8, 9] explicitly combine AQG with adaptive item delivery, and neither demonstrates a fully calibrated closed-loop pipeline.

9.2 How Our Project Addresses the Gaps

Our framework targets each identified gap directly: (1) UMLS validation filters hallucinated items before delivery (RQ1; preliminary +5 pp physician approval); (2) Bloom classification enforces cognitive-level coverage (RQ2); (3) the Haladyna auditor contributes a standardised, literature-grounded measurement protocol (Gap 3); (4) the closed-loop pipeline integrates AQG with IRT calibration and reinforcement-learning delivery (RQ3, Gap 4); (5) the proposed multi-specialty evaluation across internal medicine, pharmacology, and paediatrics targets cross-specialty generalisability (RQ4, Gap 5).

9.3 What Works

The cross-corpus distractor-mining strategy is the principal engineering result. A 66.7% functional-distractor rate sits between Tarrant 2006’s clinical-exam baseline and the human-authored MedMCQA target. The defining design choice — ranking by $\cos \times \text{len_match}$ rather than $\cos$ alone — is the v0.2 fix that simultaneously rescues the system from F7/F8 length-cue catastrophe.

9.4 What Doesn’t

NLG metrics on textbook-grounded outputs are off-distribution-suppressed (BLEU-4 of 3–4 vs lowest baseline 14.2). Fine-tuning T5 on MedMCQA train (as in our preliminary run) closes part of the gap to BLEU-4 = 16.8.

Bloom-level coverage is heavily skewed (97.8% recall). Only the reasoning-template branch produces higher-order items, and the clinical-density gate is conservative.

9.5 Limitations

- Single source PDF for textbook-grounded experiments; cross-textbook generalisation untested.
- Run B sample is small ($n=9$); per-flaw confidence intervals are wide.
- BERTScore is a TF-IDF proxy in our offline environment.
- No human-rater study on textbook-grounded outputs; preliminary 62%/67% physician approval is on MedMCQA-fine-tuned outputs only.

10 Future Work

We see five concrete next steps mapped to our RQs:

1. Complete T5-XL training with Bloom and IRT conditioning to address RQ1/RQ2.
2. Replace base T5 with ClinicalT5 or BioBART-QG; preliminary literature suggests +3–5 BERTScore.
3. Add iterative self-critique *à la* MCQG-SRefine using a small open model in place of GPT-4.
4. Knowledge-graph-guided distractors using UMLS as the KG (per Yang et al.).
5. Clinician-panel review on a 100-item sample with 30+ raters across three specialties; report Cohen’s κ for inter-rater agreement on item validity and distractor plausibility.

11 Conclusion

We present an open, lightweight, T5-based MCQ-generation pipeline whose distractor quality (66.7% functional) is in the *functional* range published in the medical-education literature, and whose surface-flaw rate after v0.2 fixes (52% flaw-free) approaches the lower band of NBME-trained item writers. Preliminary fine-tuning on MedMCQA reaches BLEU-4 = 16.8 and physician approval = 62%, rising to 67% with UMLS validation enabled.

The pipeline’s defensible engineering result is the length-matched cross-corpus mining strategy — a 10-line code change that simultaneously satisfies Tarrant’s distractor-functionality criterion and Haladyna’s F7/F8 length-cue criterion. The pattern we recommend for downstream adopters is: *measure surface-detectable IWFs on every generation run, then fix them with targeted, literature-grounded code changes*. None of our v0.2 fixes required additional model training; all are evidence-based engineering, and collectively they raised Run B’s flaw-free rate from 0% to 52%.

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Reproducibility

All code, data pre-processing scripts, and evaluation harnesses are at https://github.com/pradhyumn-nair/AQG_Generator. The MedMCQA corpus is at <https://huggingface.co/datasets/openlifescienceai/medmcqa>.

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